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Project Title:	Structure and Dynamics of Fragile Glass-Forming Systems		
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This project consists of the investigation of a series of fragile glass formers, including phosphates, chalcogenides, and aluminates, with the aim to elucidate structural developments associated with glass formation. To this end we use a combination of concurrent Raman and Brillouin light scattering as the structural probe, and molecular dynamics simulations to interpret the experimental observations. Raman scattering allows us to monitor changes in the symmetry of structural building blocks, while Brillouin scattering yields the complex mechanical modulus at the molecular scale. The storage component of this modulus provides a measure of the degree of connectivity between building blocks and the structural integrity, whereas the loss modulus assesses the energy dissipated in aperiodic motions of these building blocks. The role of simulation is to explain the observed measures in terms of structural changes that occur during glass formation.

	This methodology constitutes a new approach towards investigating glass structure and dynamics. In particular, it provides insights into the elusive intermediate-range order of non-crystalline structures. Variable parameters in this study include the nature of the network former and the chemical composition of the glass, and the preparation history, i.e., sol-gel synthesis vs. melt processing.
Task Description:	1. Devise and refine a methodology, based on concurrent Brillouin and Raman scattering, to characterize structural developments in fluids, sol-gel derived systems, and supercooled melts during the glass formation process. While we have already demonstrated that the scattering techniques can act as a structural probe on the basis of comparative evaluation, we expect that the combination with molecular dynamics simulations will allow for a direct mechanistic and structural interpretation of experimental observables. 2. Substantiate and elucidate the concept of polyamorphism, i.e., the existence of more than one thermodynamically and structurally distinct amorphous state for a given material. Specifically, we will verify whether polyamorphic transitions are ubiquitous to glass formation. 3. By understanding the fundamental principles that underlie glass formation and that control structural developments in glass-forming systems, develop knowledge that allows for a more targeted design of amorphous materials with specific properties. With this project we take a novel approach towards investigating the amorphous state of matter, and therefore expect to gain unprecedented insights in this subject matter. Accordingly, we expect to (i) advance fundamental aspects of glass science, (ii) devise knowledge for the creation of new glass-forming materials with applications in optical waveguides, photonic crystals, optical amplifiers, lasers. sensors, integrated circuits, and logic devices, and (iii) develop a technique of investigation that is of general interest for NASA's microgravity research program because it allows for the contact-free measurement of visco-elastic properties of fluids, i.e., without obliterating the environment free of external forces provided by microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While ultimately container-less melting in microgravity conditions provides the ideal environment for investigating structural developments in deeply supercooled glass-forming melts or in a sol-gel precursor maintained in delicate thermo-mechanical equilibrium, the method of investigation developed in this research can be applied in numerous situations on earth in which direct contact between specimen and the mechanical probe is difficult or impossible. Such situations include specimens that are (i) chemically aggressive, (ii) held at very high temperature, (iii) of fragile constitution (e.g., gels), (iv) of small dimensions, and (v) part of a living organism.
	While the grant for this research was awarded at the end of 2003, a graduate student qualified to work on the project could not be identified until September 2004. The project is underway at full capacity since that time. Currently, the personnel involved in the project consist of one graduate student, one part-time postdoctoral research associate, and the principal investigator. Ongoing work involves three aspects: 1. Instrumentation, experimental setup, and data analysis procedure 2. Monitoring structural developments of glasses during sol-gel synthesis using concurrent Brillouin (BLS) and Raman (RLS) light scattering 3. Molecular dynamics (MD) simulations of structural transitions in glasses and their crystalline counterparts. The progress to date is as follows: Leading up to September 2004, i.e., during the spring and summer of that year, the PI and a part-time postdoctoral research associate conducted preparatory work for the project. This consisted of working out the configuration of the Fabry-Perot interferometer, which is used for Brillouin spectroscopy and is manufactured by JSR Scientific instruments in Switzerland. The newly designed cosine interferometer is more robust and characterized by higher operating stability. Another important feature of this apparatus is that it can be outfitted with a confocal interferometer stage, which allows for higher-resolution measurements. The interferometer has been received in September 2004. It has been set up in our laboratory, aligned, and calibrated. In the spring and summer of 2005 we have worked on upgrading our Raman spectrometer. Basically, the old instrument was traded in for a newly developed triple monochromator, TR777, by Princeton Instruments – Acton Research. This instrument has a focal length of 750 mm in each of the three monochromators and can be operated in subtractive or additive mode. This provides superb laser line suppression and high spectral resolution. The instrument is equipped with a Spec10:400B LN2-cooled CCD detector, which has a 1340 by 400 pixel resolution and can be operated in image mode. While this instrument has been on the market for three years in Europe, ours is the first one installed in North America. Meanwhile the graduate student working on the project has been trained in using both of these instruments. He has also familiarized himself with sample preparation techniques and is now working to prepare chalcogenide and phosphate glasses. The student has developed remarkable proficiency in the synthesis of glasses using the sol-gel route. In addition, the student and the part-time postdoctoral associate have developed data analysis software. This software is specifically designed for the interpretation of the Brillouin spectra in terms of the generalized hydrodynamic theory, i.e., to extract both elastic moduli and viscosity coefficients from this data. The analysis procedure involves deconvolution to rid the spectra of instrument broadening and the fitting of a theoretical line shape to the deconvoluted data. The software is designed to run on unix platforms in an X-window environment. Finally, we are currently working on a new design for a high-temperature sample cell. (Our existing high-temperature cell is quite old. The platinum heating elements are severely recrystallized, i.e., they are brittle and their Ohmic resistance is getting too low to be accommodated by the power supply unit.) The new design will be more compact, and capable of multiple scattering angles. Initially, we focused our experimental investigation on phosphate glasses, specifically, a series of glass compositions in the SiO2-P2O5 system. This course of action was indicated for the following arguments. (1) Based on the current literature, the standard procedure for preparing phosphate and phospho-silicate glasses is the sol-gel method, followed by

Task Progress:	varying degrees of heat treatments. Furthermore, given that phosphate-rich compositions have relatively low glass transition temperatures, this system is ideally suitable for carrying out the proposed comparative investigation of glasses prepared via sol-gel synthesis and conventional melt processing. (2) Phospho-silicate glasses have recently received significant attention for potential application as the electrolyte material for proton exchange membranes. These glasses exhibit the highest proton conductivity of any all-oxide material. This is due to the superb protogenic properties of phosponic groups and to the fact that these glasses form a unique structure, in which the silica and phosphorous pentoxide components are phase separated at the nano-scale. The two phases arrange into two interpenetrating networks; the silica-rich one provides mechanical stability and the phosphate-rich one provides percolating paths for proton transport. (3) The precise nature of the structures that develop in these systems is strongly dependent on the processing parameters and heat treatment schedules. This system should therefore be very rich in terms of structural phenomena to be examined in the context of polyamorphism, and that should lead to a better understanding of this concept. Using concurrent Brillouin and Raman scattering, we have to date (1) carried out a systematic study of the gelation process, (2) monitored structural developments of the sol-gel derived materials upon drying, between gelation point and fully dense xerogel, and (3) done both of these studies as a function of the chemical composition of the materials within the SiO2-P2O5 system. The most noteworthy results are as follows. 1. We observe that the visco-elastic properties of aclogels and xerogels initially decrease slightly and then considerably increase with increasing phosphorous pentoxide concentration. This pertains to both the elastic modulus (given by the frequency shift in the Brillouin spectrum) and the viscosity coefficient (given by the width at half maximum of the Brillouin peaks). This is to some degree surprising, as the stronger silica network becomes increasingly disrupted with the addition of phosphorous pentoxide. 2. Elastic moduli and viscosity coefficients track none another, i.e., they undergo commensurate changes. This can be explained by the fact that upon drying the number of cross links in the networks increases, which provides more mechanical rigidity, but also constrains the fluid that flows across smaller and smaller pores. 3. The gelation point in all systems is characterized by a ubiquitous cusp in both the frequency shift (elastic modulus) and width (viscosity). Accordingly, the viscosity of the solution increases gradually until gelation occurs. Immediately following gelation, the viscosity decreases slightly, or at least remains stagnant for some time while the gel continues to densify through syneresis and drying, before it finally increases as the gels continue to shrink. It is not entirely clear what causes this behavior. A possible explanation is that long-range Coulomb interactions prevail in the system and affect fluid flow behavior up to the gelation point. At this point a significant number of covalent bonds are formed between the structural building blocks, which is accompanied by a charge redistribution and neutralization of the long-range interactions. This in turn relaxes the mechanical constraints on the system, which is reflected in lower elastic moduli and viscosity coefficients. 4. Interestingly, as is apparent from Raman spectra, no significant structural change in the near-range structure takes place at the gelation point, i.e., the geometry of the basic structural units does not seem to be affected by the sudden combination of building blocks into a three-dimensional network. The most significant changes in Raman spectra occur during the later stages of the densification process. On the computational side of the project, which was carried out by the part-time postdoctoral associate, we have continued to improve our interaction potential, specifically in its ability to simulate systems that exhibit variable coordination (e.g., 4- or 6-coordinated Si and 3- or 4-coordinated B). Using simulations based on this potential we were able to: 1. Predict new high-pressure crystalline phases in silica, and discover a ubiquitous transformation path for both quartz and cristobalite. This densification mechanism involves an intermediate step of forming a close-packed oxygen sublattice, which in case of cristobalite corresponds to a metastable phase (X-I). 2. Discover two new low-density crystalline phases in boron oxide. The phases form under isotropic tensile stress, which is difficult to realize in experiments. They also manifest themselves as a short-lived transitional state upon heating the known high-density modifications of boron oxide, just before melting occurs. The significance of the low-density phase is explained in item 3. 3. Discover an anomalous minimum in the elastic modulus of vitreous boron oxide under tension. This is a behavior similar to that observed in silica under pressure, and can be explained in analogous terms as due to transitions between two types of ring structures, one of them resembling those in a low-modulus high-density (alpha) crystalline phase, and the other one those in a high-modulus low-density (beta) phase. This discovery reveals a generalization of the origin for such anomalous thermo-mechanical properties in network glasses, which is rooted in polyamorphic transitions.

Bibliography Type:	Description: (Last Updated: 02/26/2014)
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